

KATRAVATH SRINU

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EDUCATION

Indian Institute of Technology (BHU), Varanasi

B.Tech in Computer Science and Engineering

CGPA: 6.75/10

Expected Graduation: 2027

Relevant Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Probability & Statistics, AI, Machine Learning, Discrete Mathematics

PROJECTS

Netflix Recommendation System (Netflix Prize-Style)

Python, NumPy, Pandas, Scikit-learn

- Built a **Netflix-style recommendation system** leveraging **SVD-based collaborative filtering and latent factor models** on large-scale user-item rating data.
- Optimized model performance by minimizing **RMSE**, improving recommendation accuracy over baseline approaches while efficiently handling sparse datasets.
- Designed end-to-end data preprocessing and evaluation pipelines, including normalization, train-test splitting, and performance analysis.

AI Resume Screening Tool

Python, NLP

- Developed an **NLP-based resume screening system** to parse PDF resumes and extract key candidate skills automatically.
- Implemented a **similarity-scoring algorithm** to rank candidates against job descriptions, enabling **automated and objective candidate shortlisting**.

Real-Time Chat Application

Node.js / Java, WebSockets, MongoDB

- Developed a **scalable real-time chat application** supporting one-to-one and group messaging using **WebSockets**.
- Implemented **message delivery acknowledgments** and robust online/offline user state handling.
- Designed backend APIs to manage users, chat rooms, and persistent message storage.

SKILLS

- **Programming:** C++, Python, C, SQL
- **CS Fundamentals:** Data Structures & Algorithms, OOP, DBMS, Operating Systems.
- **Machine Learning:** Collaborative Filtering, Matrix Factorization (SVD), Model Evaluation, NLP Techniques.
- **Web / Backend:** REST APIs, Flask, WebSockets.
- **Tools:** Git, GitHub, Linux, VS Code.

ACHIEVEMENTS

- Solved **250+ DSA problems** across LeetCode and Codeforces (**Codeforces Max Rating: 1100**).
- **JEE Advanced 2023:** AIR **11227** among ~1.5 lakh candidates.
- Selected for **B.Tech CSE at IIT (BHU)** through JEE Advanced.

INTERESTS

- Machine Learning Systems, Competitive Programming, Backend Engineering, System Debugging.